

Project Title: QueueBank MS / Smart Bank Appointment Reservation System

Course: Software Engineering

Methodology: DMAIC Methodology Project

Phase: Phase III — Software Design and Modeling

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System Architecture

Customer Books an Appointment:

- Customer selects a service, date, and time slot (Frontend — user.html)
- Request goes to Backend (add_booking.php)
- Backend checks the Bookings table in the Database to see if the selected time slot is already taken
- Backend saves the new booking into the Bookings table and generates a unique QB-XXXXX confirmation code
- Backend sends the confirmation code back to Frontend
- Customer sees the confirmation screen with the code (Frontend — user.html)

Staff Manages Appointments:

- Staff enters Employee ID and password (Frontend — staff.html)
- Request goes to Backend (staff_login.php)
- Backend checks the Staff table in the Database to validate the credentials
- Backend sends the appointments list back to Frontend from the Bookings table
- Staff marks an appointment as Completed or No-show (Frontend — staff.html)
- Request goes to Backend (update_status.php)
- Backend updates the status of that booking in the Bookings table in the Database

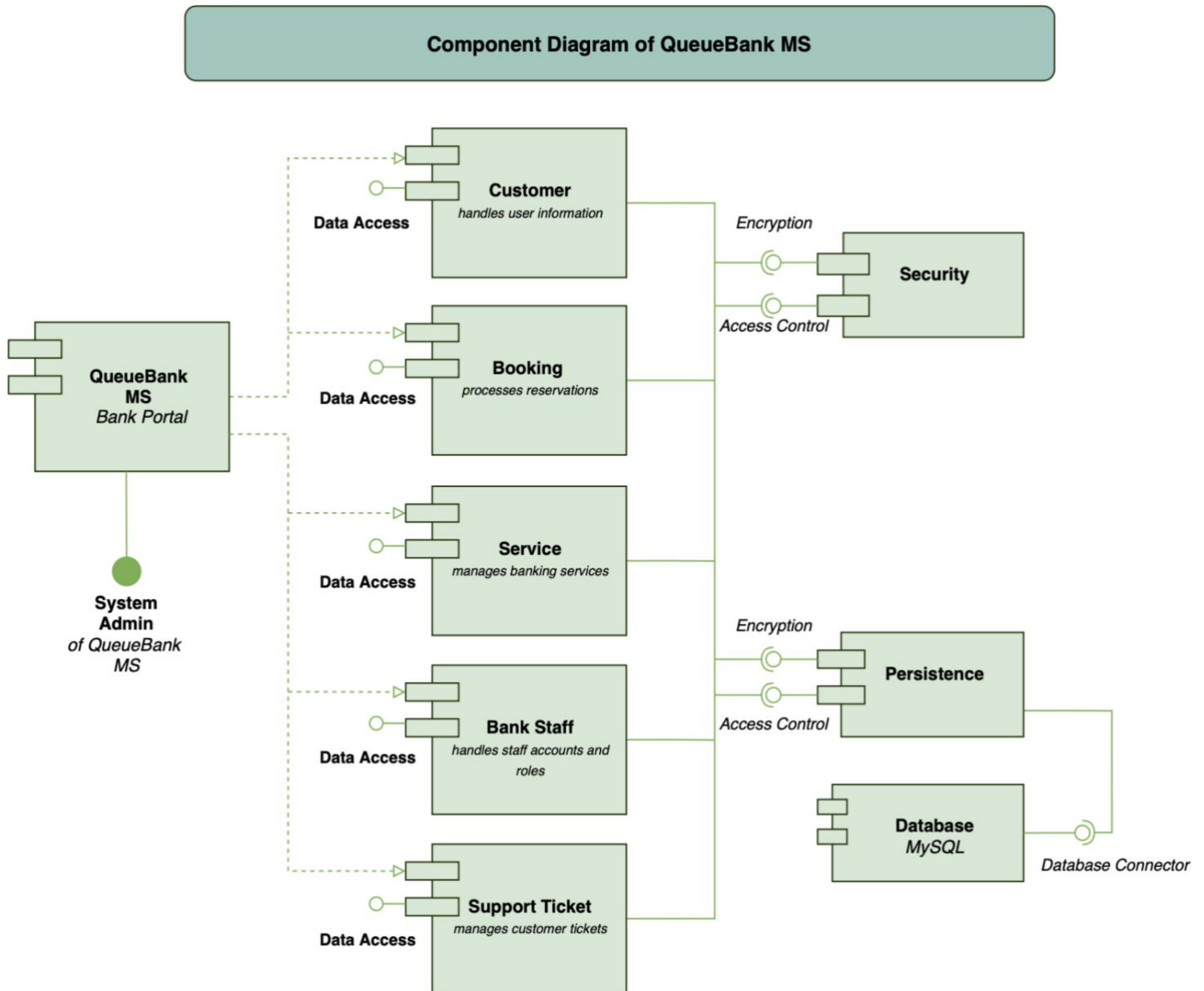
Customer Submits a Support Ticket:

- Customer fills in the subject and message (Frontend — user.html)
- Request goes to Backend (add_ticket.php)
- Backend validates the form and saves the ticket into the Support Tickets table in the Database
- Backend sends a success confirmation back to Frontend
- Customer sees a success message (Frontend — user.html)
- Admin opens the Support Tickets section (Frontend — admin.html)
- Request goes to Backend (get_tickets.php)
- Backend reads all tickets from the Support Tickets table in the Database
- Backend sends the tickets list back to Frontend
- Admin sees all submitted tickets and can update their status (Frontend — admin.html)

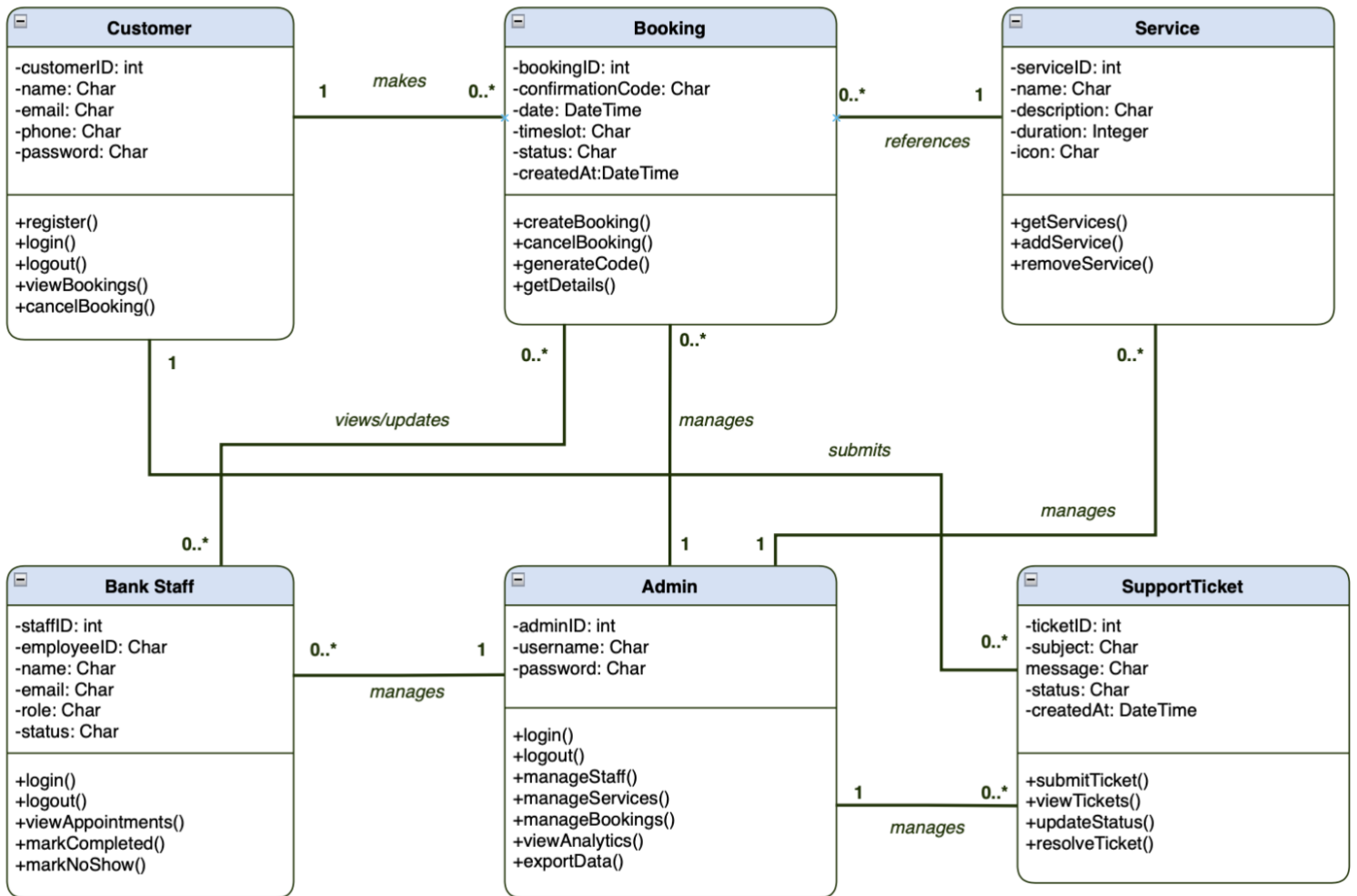
Admin Manages the System:

- Admin enters username and password (Frontend — admin.html)
- Request goes to Backend (admin_login.php)
- Backend checks the Admin table in the Database to validate the credentials
- Admin can add or remove staff — Backend updates the Staff table in the Database
- Admin can add or remove services — Backend updates the Services table in the Database
- Admin can view booking analytics — Backend reads from the Bookings table and sends data back to Frontend

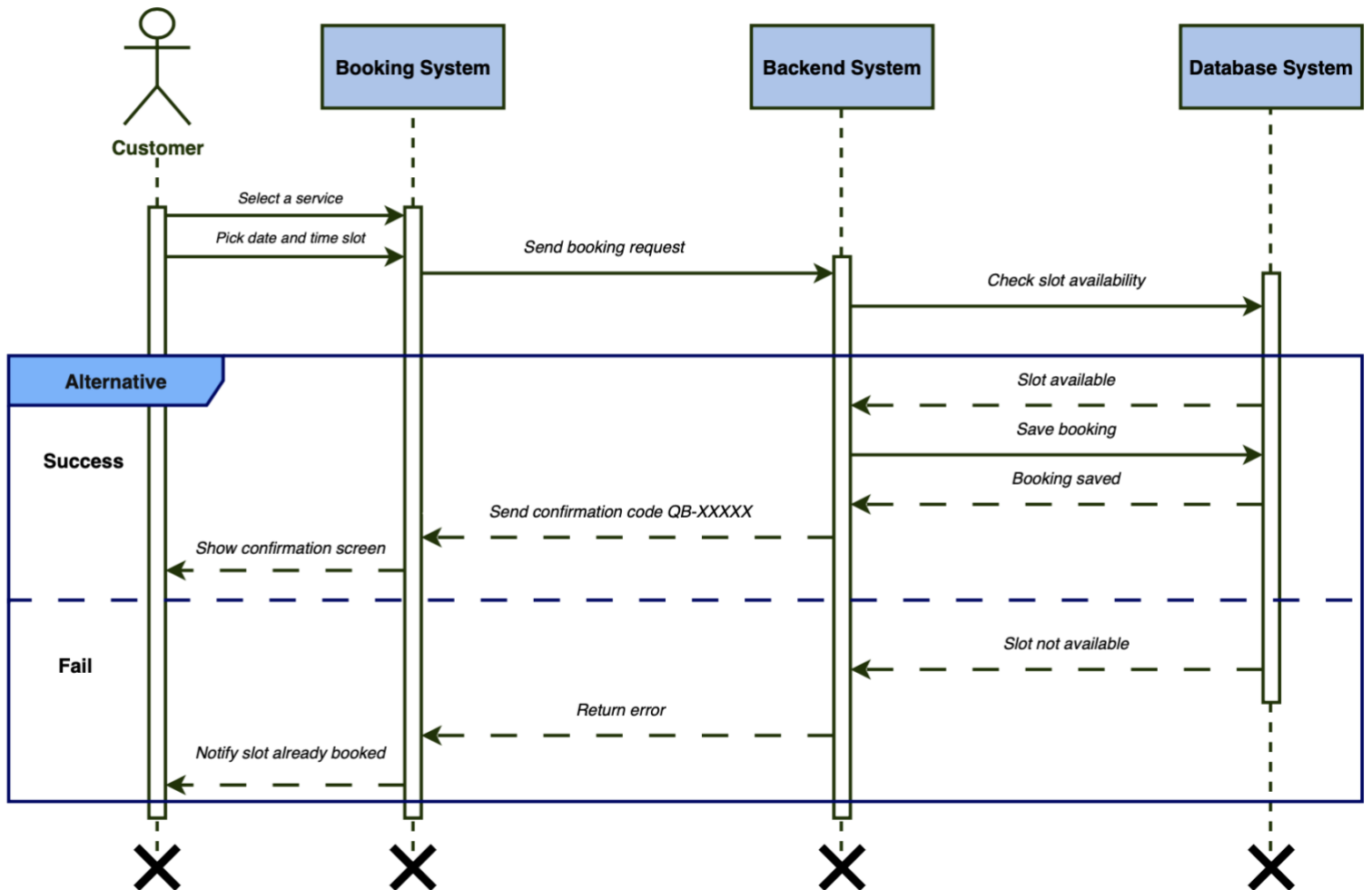
- Admin can export all data — Backend reads all tables and returns a JSON file to Frontend



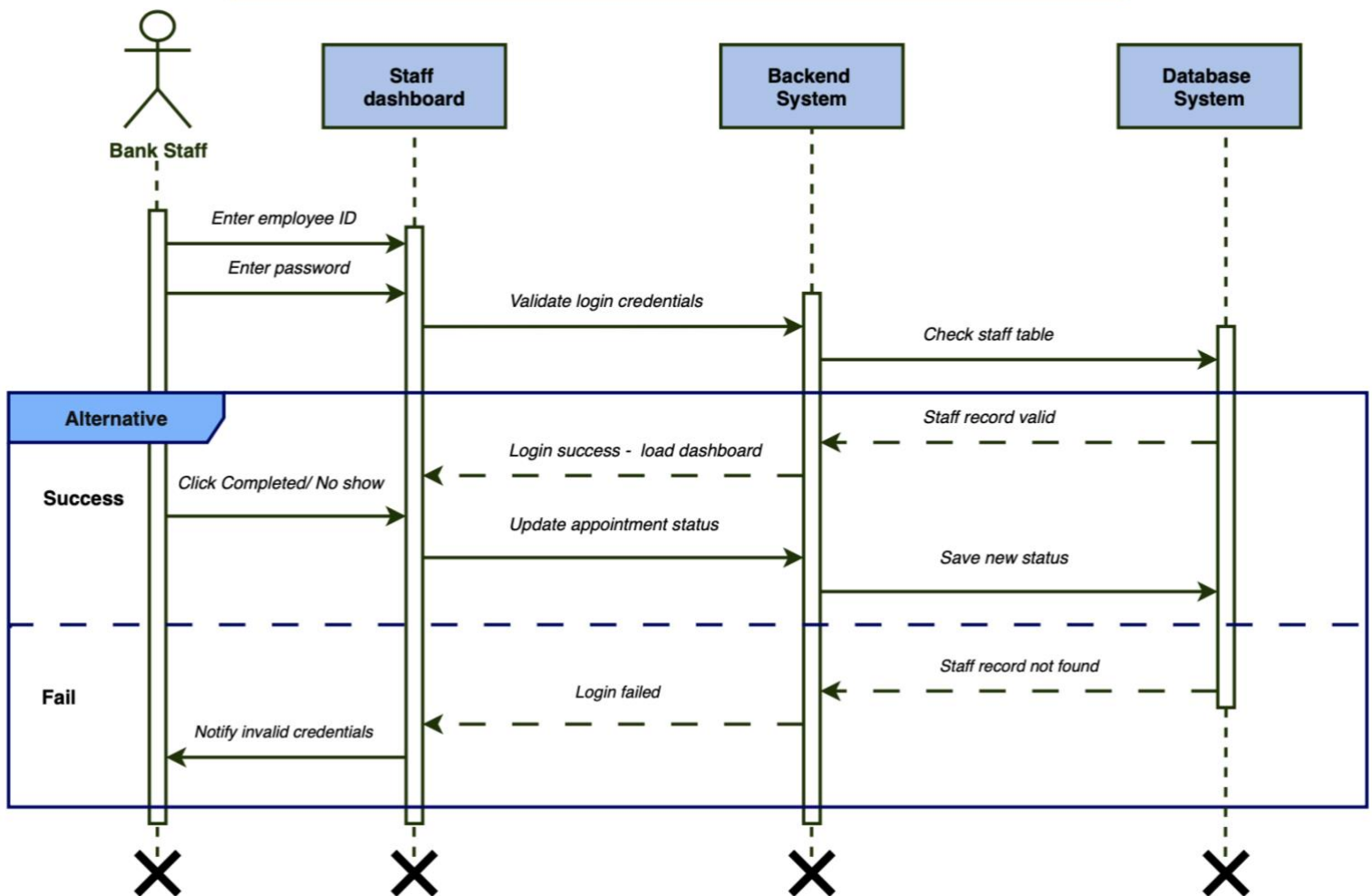
The Class Diagram of QueueBank MS



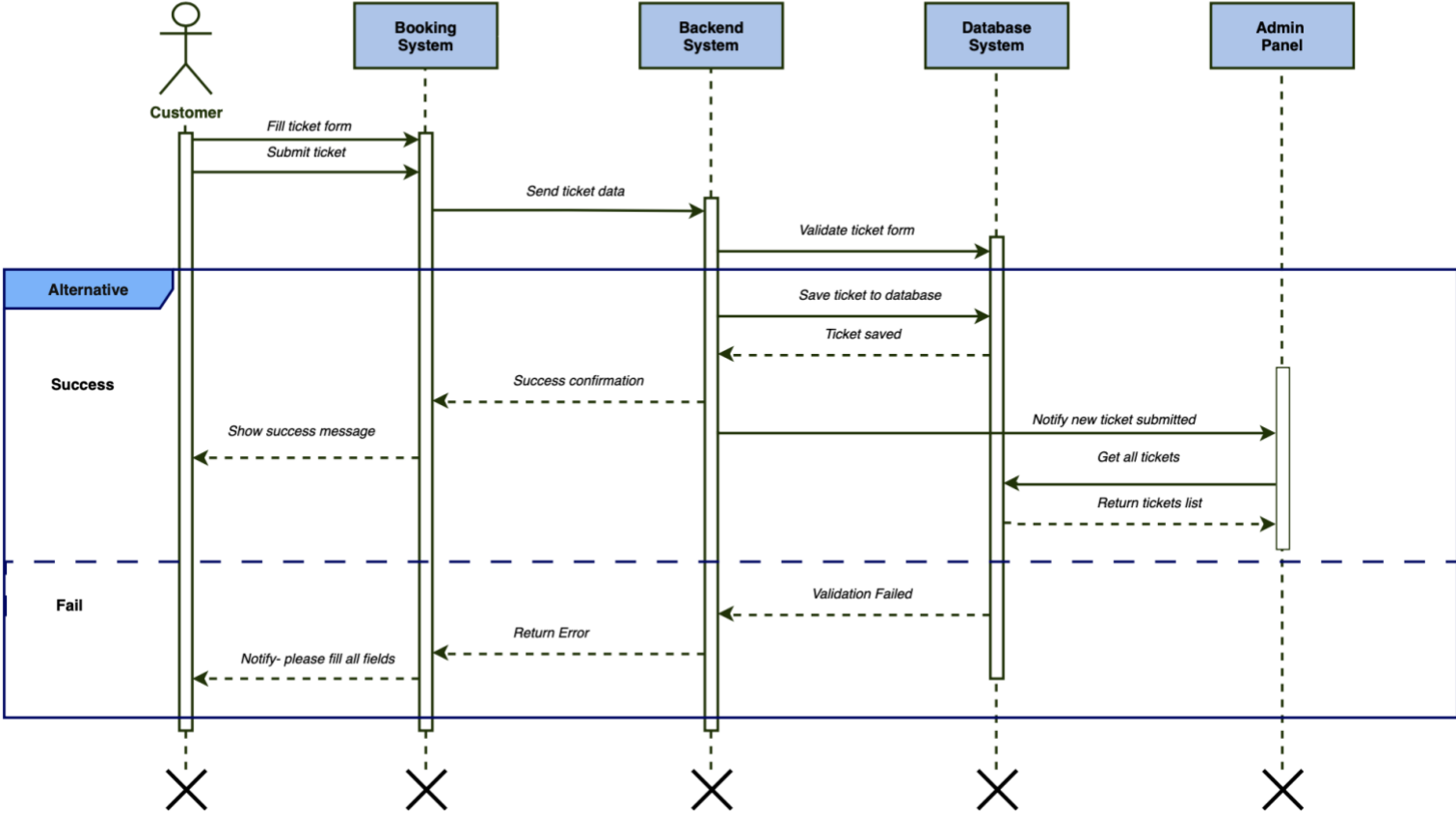
Sequence diagram 1 - Book an Appointment



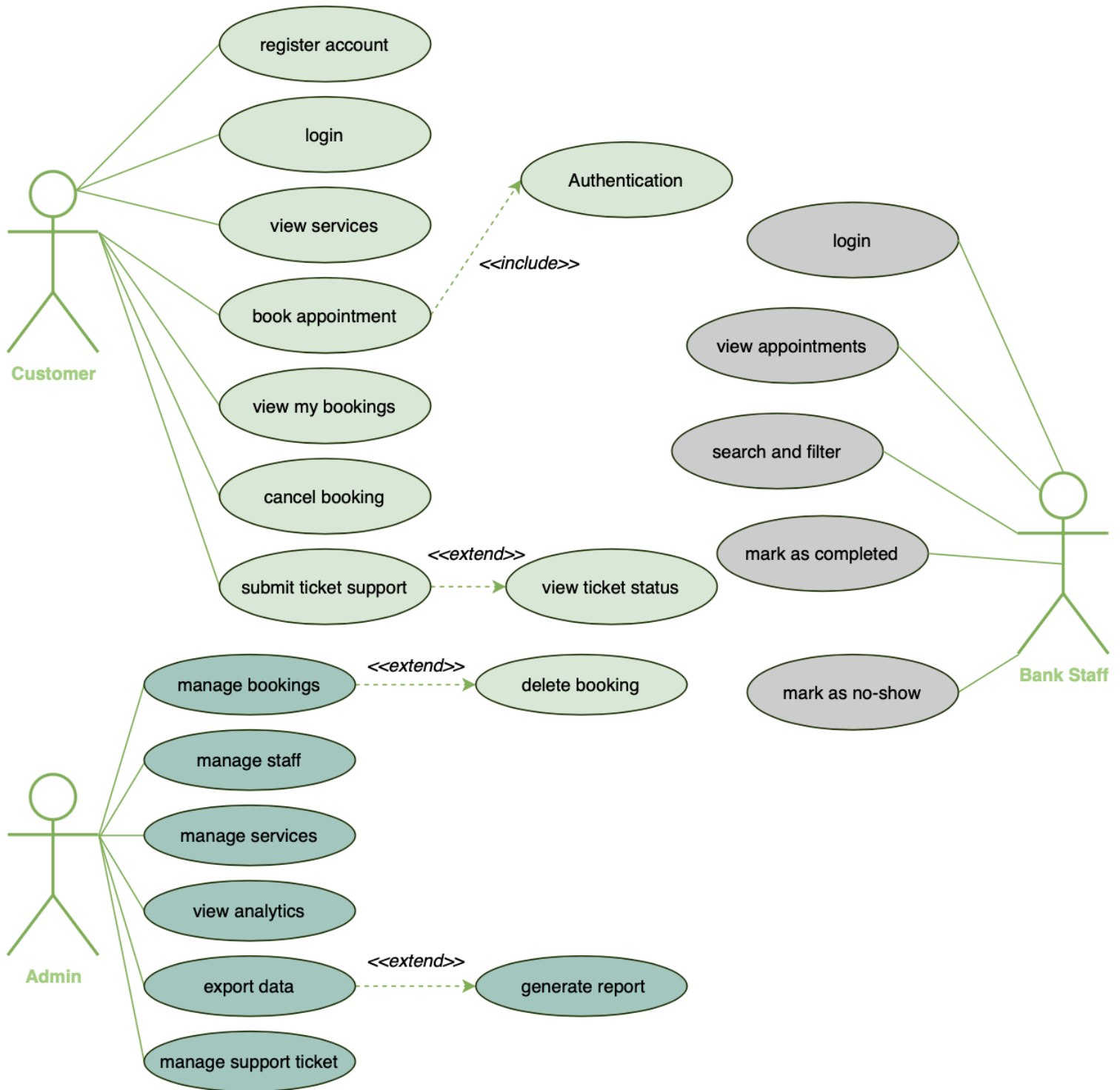
Sequence diagram 2 - Staff Marks Appointment



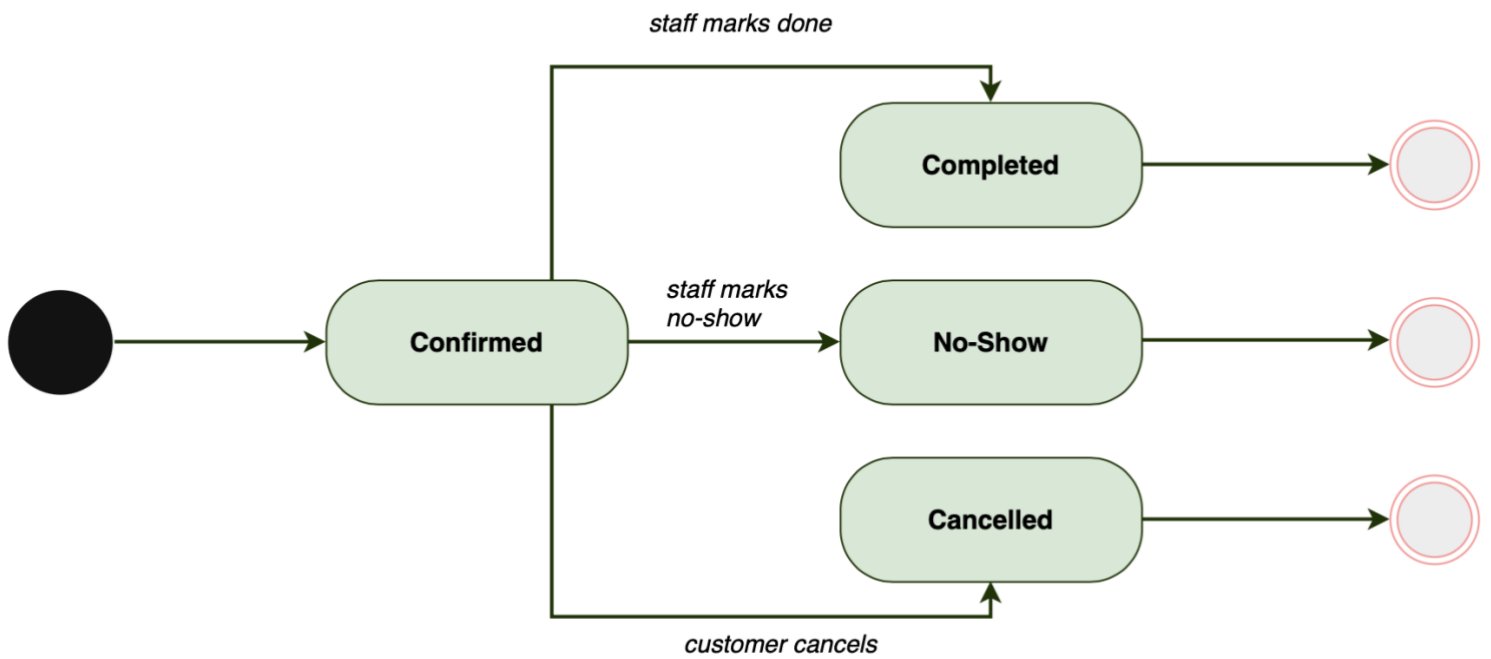
Sequence diagram 3 - Customer Submits Support Ticket



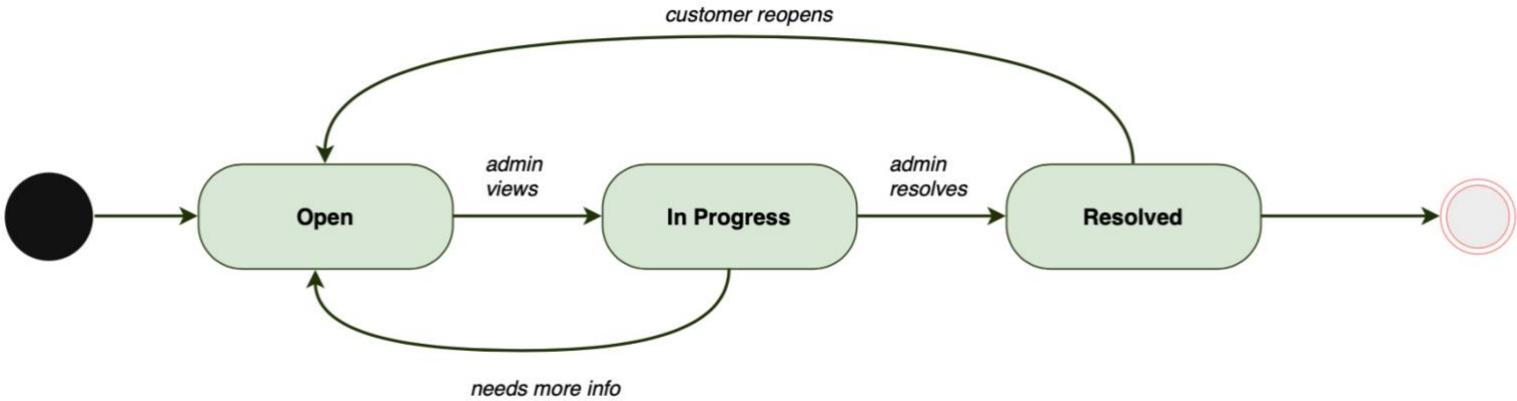
QueueBank MS - Use case Diagram



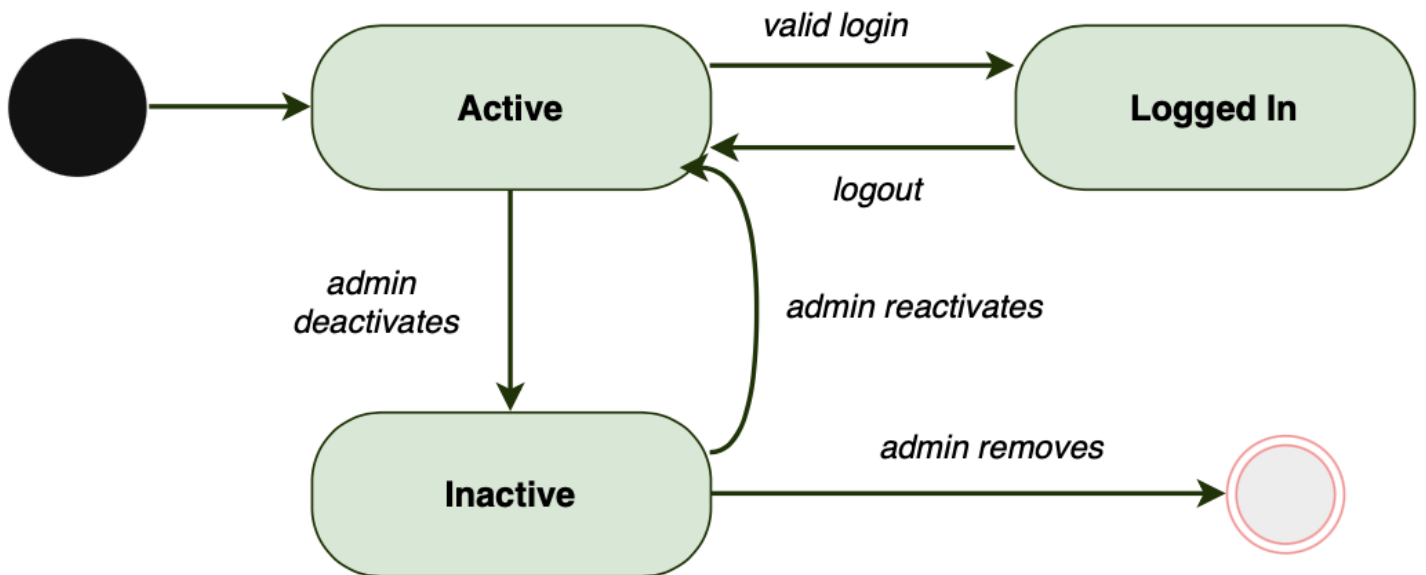
State diagram 1 - Booking Status



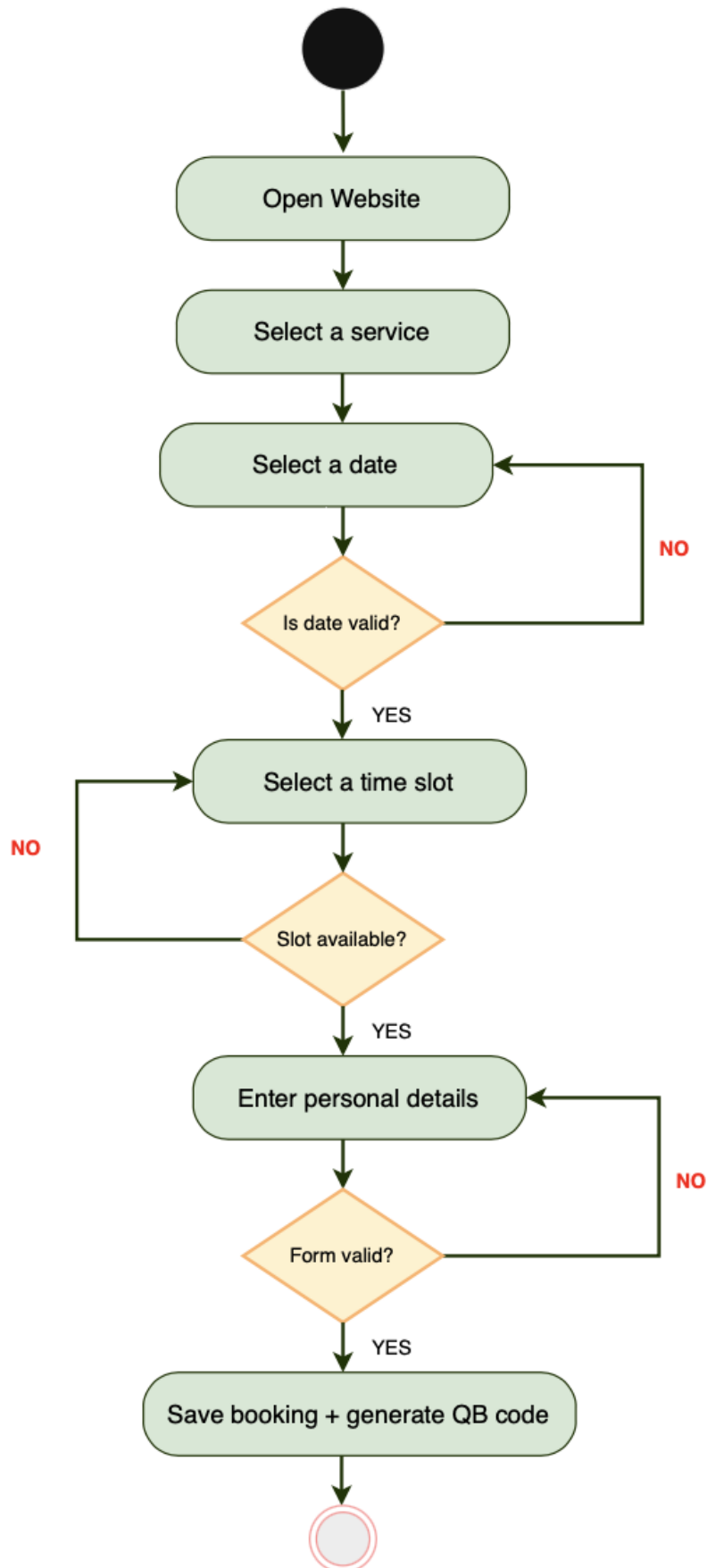
State diagram 2 - Support Ticket Status



State diagram 3 - Staff account Status



Activity diagram 1 - Book appointment



Activity diagram 2 - Staff Manages Appointments

